Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$a = 28$
			Total 1 mark

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Shows integral of v with respect to t Condone omission of dt	AO1.1a	M1	$\int 0.06(2 + t - t^2)dt$
	Integrates with at least two correct terms (condone omission of constant at this stage) Any equivalent form.	AO1.1b	A1	$= 0.12t + 0.03t^2 - 0.02t^3 + c$
	Finds constant using $t = 0$ and displacement of 3. PI.	AO2.2a	M1	Using $t = 0$, $c = 3$
	Finds fully correct expression for r.	AO1.1b	A1	$r = 0.12t + 0.03t^2 - 0.02t^3 + 3$
(b)	Models problem as only force acting is that due to gravity so that $v = 0$ at highest point. (PI)	AO3.3	M1	$a = -9.8$ m s^{-2} $v = 0$ m s^{-1}
	Uses appropriate suvat formula $v = u + at$ with $u = 3.43$ and g negative or $u = -3.43$ and g positive	AO1.1a	M1	$0 = 3.43 - 9.8t_{max}$
	Finds $t_{max} = 0.35$ s (CAO) Condone addition of the 2 seconds to give $t = 2.35$ s or 2.4 s	AO1.1b	A1	$\therefore t_{max} = 0.35$ s
				Total 7 marks

Q3.

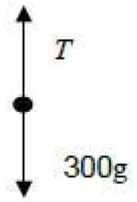
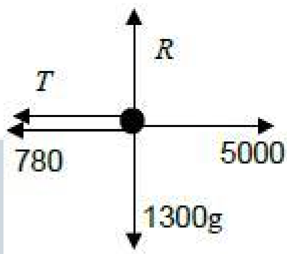
	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses model for maximum friction = μmg	AO3.3	B1	$F_{\text{max}} = \mu mg$ $= 0.85 \times 20 \times 9.8$ $= 166.6$ N
	Makes an appropriate comparison	AO1.1a	M1	
	Explains clearly why crate remains stationary	AO2.4	E1	150 < 166.6 ∴ crate does not move
(b)	Forms an equation by resolving vertically Condone one of sign error or cos error	AO3.1b	M1	$20g = R + 150\sin 15°$ $R = 157.177$ N
	Obtains correct reaction force	AO1.1b	A1	$F_{\text{max}} = \mu \times 157.177$ $= 133.6$ N
	Uses maximum friction = μR With 'their' reaction force Must identify maximum or limiting friction	AO1.1b	B1F	$150\cos 15° = 145$ N 145 > 133.6 ∴ crate begins to move
	Compares 150cos15° with 'their' maximum friction	AO1.1a	M1	
	Explains, using their values, why the crate begins to move.	AO2.4	E1F	
				Total 8 marks

Q4.

Marking Instructions	AO	Marks	Typical Solution
Applies Newton's 2nd Law to form a 3 term equation Award mark even if signs not correct	AO1.1a	M1	$F - 80 \times 10 = -80 \times 1.5$
Obtains a correct 3 term equation.	AO1.1b	A1	$F - 800 = -120$
Obtains correct reaction force. Must be given to 1 sf FT from incorrect 3 term equation provided M1 mark was awarded (condone omission of	AO1.1b	A1F	$F = 680 = 700$(N) to 1 sf

units)			
			Total 3 marks

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Draws correct force diagram for crate from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	
(ii)	Draws correct force diagram for van from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	
(b)	Applies Newton's 2nd Law ($F = ma$) to the crate	AO3.4	M1	For crate $T - 300\text{g} = 300a$
	Applies Newton's 2nd Law ($F = ma$) to the van ($F = ma$ 'round the corner' scores 0)	AO3.4	M1	For van $5000 - T - 780 = 1300a$ $4220 - T = 1300a$
	Solve their simultaneous equations	AO1.1a	M1	$4220 - 300\text{g} = 1600a$
	Finds the value of a correctly **AG**	AO1.1b	A1	$a = 1280 \div 1600 = 0.80$ m s $^{-2}$ **(AG)**
(c)	Uses $a = 0.80$ in either of their two equations in **(b)**	AO3.4	M1	$T = 300 \times 0.80 + 300\text{g}$
	Finds the correct value for T (condone omission of units) Possibly done in **(b)**	AO1.1b	A1	= 3180 = 3200 (N) (2 sf)
(d)	Explains that the model could be refined by including air resistance	AO3.5c	E1	Resistance will increase with speed

Total 9 marks

Q6.

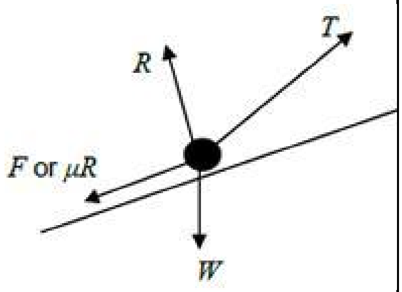
Marking Instructions	AO	Marks	Typical Solution
Uses correct forces to form a moment equation **(PI)**	AO1.1a	M1	Take moments about C: Mg × 0.8 = 0.7 × 24
Obtains correct value	AO1.1b	A1	M = 21
			Total 2 marks

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Integrates both components with at least one correct	AO1.1a	M1	$\mathbf{r} = \int 40\mathrm{e}^{-0.2t}\,\mathrm{d}t\,\mathbf{i} + \int 50(\mathrm{e}^{-0.2t} - 1)\,\mathrm{d}t\,\mathbf{j}$ $= (-200\mathrm{e}^{-0.2t} + c)\mathbf{i} + (-250\mathrm{e}^{-0.2t} - 50t + d)\mathbf{j}$
	Obtains correct terms. (condone missing constants)	AO1.1b	A1	t = 0, **r** = 0**i** + 0**j** ⇒ c = 200, d = 250 **OR**
	Evaluates both constants (or uses definite integration) using 'their' expression for **r**	AO3.4	M1	$\mathbf{r} = \int_0^t 40\mathrm{e}^{-0.2t}\,\mathrm{d}t\,\mathbf{i} + \int_0^t 50(\mathrm{e}^{-0.2t} - 1)\,\mathrm{d}t\,\mathbf{j}$ $= \left[-200\mathrm{e}^{-0.2t}\mathbf{i} + (-250\mathrm{e}^{-0.2t} - 50t)\mathbf{j}\right]_0^t$
	Obtains correct expression	AO1.1b	A1	**r** = 200(1 − $\mathrm{e}^{-0.2t}$)**i** + (250 − 250$\mathrm{e}^{-0.2t}$ − 50t)**j**
(b)	Forms equation to find t based on horizontal component	AO3.4	M1	$200(1-\mathrm{e}^{-0.2t}) = 100$ $t = 5\ln 2 = 3.4657$ $y = 250 - 250\mathrm{e}^{-0.2\times 5\ln 2} - 50\times 5\ln 2$ $= -48.3$
	Obtains correct time	AO1.1b	A1	
	Substitutes 'their' time into vertical component	AO1.1a	M1	
	Obtains correct displacement for 'their' time (to 1 sf, must have metres) FT only if both M1 marks have been awarded	AO3.2a	A1F	The parachutist has a vertical displacement of 50 m below the origin

	Marking Instructions	AO	Marks	Typical Solution
(c)	Identifies vertical component of the velocity as first step	AO2.4	M1	$\frac{d}{dt}(50(e^{-0.2t} - 1)) = -10e^{-0.2t}$ As there is no initial vertical component of velocity (and hence no air resistance) the initial acceleration is only due to gravity. Hence g is taken as 10 m s^{-2}
	Differentiates vertical component of velocity correctly	AO1.1b	A1	
	Considers implication of initial motion and reaches correct conclusion	AO2.2a	R1	
				Total 11 marks

Q8.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves horizontally and vertically to obtain expressions for F and R (allow consistent mixing of sin and cos)	AO3.4	M1	Resolving vertically $R = 8 \times 9.8 - 50\sin 40°$ Resolving horizontally $F = 50\cos 40°$
	Obtains correct expressions for R and F	AO1.1b	A1	
	States friction model $F = \mu R$ with 'their' values for F and R	AO1.2	B1	$F = \mu R$ $50\cos 40° = \mu(8 \times 9.8 - 50\sin 40°)$ $\mu = \frac{50\cos 40°}{8 \times 9.8 - 50\sin 40°}$
	Completes a rigorous argument that results with correct μ **AG** Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$\mu = 0.8279660445 = 0.83$ (2sf) **AG**
(b)(i)	Draws correct diagram with exactly four forces showing arrow heads and labels Can use Mg or $8g$ or 78.4 for W	AO3.3	B1	

(ii)	Resolves perpendicular to the plane resulting in a three term equation containing; R, $8g\cos 5°$ and $T\sin 40°$ (or $T\cos 50°$) OR Resolves horizontally and vertically to obtain equations of motion in horizontal and vertical directions	AO3.1b	M1	$R = 8 \times 9.8\cos 5° - T\sin 40°$ $= 78.1 - T\sin 40°$
	Obtains correct expression for R OR Obtains correct horizontal and vertical equations	AO1.1b	A1	$T\cos 40° - 8 \times 9.8\sin 5° - F = 8 \times 3$ $T\cos 40° - 8 \times 9.8\sin 5° - 0.827\ldots \times (8 \times 9.8\cos 5° - T\sin 40°) = 24$
	Forms a four term equation of motion parallel to the plane with correct terms (allow sign errors) OR Eliminates R to solve for T	AO3.1b	M1	$T = \dfrac{24 + 8 \times 9.8\sin 5° + 0.827\ldots \times 8 \times 9.8\cos 5°}{\cos 40° + 0.827\ldots \times \sin 40°}$ $T = 73.55939193 = 74$ (2 sf)
	Obtains correct equation of motion OR Obtains correct expression(s) without R	AO1.1b	A1	
	Uses the friction model in the four term equation of motion where R is in the form $a - bT$ and a and b are positive constants OR Uses the friction model in the horizontal and vertical equations Dependent on previous M1	AO3.1b	dM1	
	Solves for T	AO1.1b	A1	
	Obtains correct value of T with 2 sf accuracy. FT incorrect value found for T provided both M1 marks and dM1 mark have been awarded	AO3.2a	A1F	

Total 12 marks

Q9.

(a) $3g - T = 3a$

M1: Three term equation of motion with $3g$ or 29.4, T and $3a$.
A1: Correct equation.

M1A1

$T - g = a$

$2g = 4a$

M1: Three term equation of motion with g or 9.8, T and a.
A1: Correct equation.

M1A1

$a = \frac{g}{2} = 4.9 \text{ m s}^{-2}$

A1: Correct final answer. Accept $\frac{g}{2}$

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they then give their final answer as 4.9, having seen –4.9 in their working. If the final answer is –4.9 don't award the final A1 mark.
Special Case:

Whole string method $2g = 4a$ and $a = \frac{2g}{4} = 4.9$ OE scores M1A1A1

5

(b) $v^2 = 0^2 + 2 \times 4.9 \times 0.4$

M1: Use of a constant acceleration equation to find v, with $u = 0$, their value for a from part (a) and $s = 0.4$ or 40.

M1

$v = \sqrt{3.92} = 1.98 \text{ m s}^{-1}$

A1F: Correct speed. Follow through their acceleration from part (a).
Use $v = \sqrt{0.8a}$ for FT.
Accept 1.97.
If candidates use two equations, award no marks until they

have an equation for v. (Note use of t = 0.404 or better required for A1)

A1F

2

(c) $0^2 = \left(\sqrt{3.92}\right)^2 + 2 \times (-9.8)\, s$

M1: Use of a constant acceleration equation with $v = 0$, $a = \pm 9.8$ and their speed from (b).

M1

$s = \dfrac{3.92}{2 \times 9.8} = 0.2$ m

A1: Correct distance.

A1

Total = 0.2 + 0.4 = 0.6 m

A1: Correct total distance. Allow 60 cm from correct working.
Note

$0^2 = \left(\sqrt{392}\right)^2 + 2 \times (-9.8)s$

$s = \dfrac{392}{2 \times 9.8} = 20$

scores M1A0A0

A1

If candidates use two equations, award no marks until they have an equation for s. (Note use of t = 0.202 or better required for A marks)

3

(d) The acceleration would be less, because the <u>resultant force</u> on each particle would be <u>reduced</u>.

B1: Less
'Slower acceleration' not acceptable

B1

B1: Appropriate reason.
Only award second B1 if they say acceleration is less.

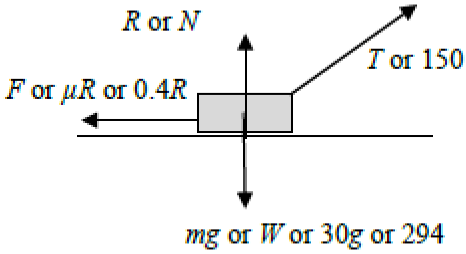
B1

2

[12]

Q10.

(a) (i)



B2: Correct diagram with exactly four forces showing arrow heads and labelled.
B1: Diagram with one error or omission.
B0: Diagram with 2 or more errors or omissions.

If components are also shown and they use a different style, eg dashed lines, they can be ignored.

If both components are shown in the same style as other forces, this counts as two errors.

Note; Do not accept 30kg for the weight.

B1

B1

2

(ii) $R + 150 \sin 20° = 30 \times 9.8$

$(R =)\ 30 \times 9.8 - 150 \sin 20°$

$= 242.69……$

M1: Resolving vertically to obtain a three term equation, with R, 150 sin or cos(20° or 70°) and $30g$ oe.

A1: Correct equation. Allow g instead of 9.8.

M1A1

$= 243$ N (to 3sf)

A1: AG Correct final answer having seen either 2nd or 3rd or both line of solution.

A1

3

(iii) $(F =)\ 0.4 \times 242.7 = 97.1$ N

M1: Use of $F = \mu R$ or $F \le \mu R$.
A1: Correct final answer without an inequality.
Accept 97.2.

M1A1

2

(iv) $30a = 150 \cos 20° - 97.08$

M1: Three term equation of motion with $30a$, 150 sin or

cos(20 or 70°) and their friction from (a)(iii). Condone incorrect signs.
A1: Correct equation.

M1A1

$$a = \frac{150\cos 20° - 97.08}{30} = 1.46 \text{ m s}^{-2}$$

dM1: Solving for a.

dM1

A1: Correct acceleration. Accept 1.45 or 1.47 or AWRT 1.46.

A1

4

(b) $R = 30 \times 9.8 - T\sin 20°$

B1: Correct normal reaction in terms of T.

B1

$F = 0.4\,(30 \times 9.8 - T\sin 20°)$

B1: Correct friction in terms of T

B1

$T\cos 20° = 0.4\,(30 \times 9.8 - T\sin 20°)$

M1: Resolving tension horizontally and equating to F, provided that F is in terms of T.
A1: Correct equation.

M1A1

$$T = \frac{0.4 \times 30 \times 9.8}{\cos 20° + 0.4\sin 20°} = 109 \text{ N}$$

A1: Correct tension. AWRT 109.

A1

5

(c) The same

B1: The same.

Use of g = 9.81 gives acceptable final answers.

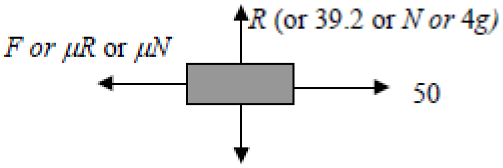
B1

1

[17]

Q11.

(a)



mg or W or $4g$ or 39.2

B1: Correct force diagram with four forces with arrows and labels. Accept words eg friction instead of letters.
Ignore negative signs in labels.
Do not accept 4 kg for the weight.
Award marks if forces are drawn on the diagram in the question.

B1

1

(b) 39.2 N

B1: Correct reaction force.
Accept 4g. Do not accept 39.

B1

1

(c) $50 - F = 4 \times 3$

M1: Three term equation of motion with the correct terms.
A1: Correct equation with correct signs.

M1A1

$F = 38$

A1: Correct friction.

A1

3

(d) $38 = \mu \times 39.2$

M1: Use of $F = \mu R$ with their answers to (b) and (c).

M1

$\mu = \dfrac{38}{39.2} = 0.969$

A1F: Correct μ based on their answers to (b) and (c).
Accept AWRT 0.969.

A1F

Note: $F = 12$ leads to 0.306 and award M1 A1F

Condone 0.97 or FT to 2sf

Condone use of inequalities.

2

(e) Less friction, so a smaller coefficient of friction.

B1: Less friction.

B1

B1: Smaller μ.

B1

Note:

More friction anywhere scores B0 B0

Less friction, greater μ scores B1 B0

Smaller μ with no / inexact reason B0 B1

2

[9]

Q12.

(a) 20 cosθ = 10

M1: Resolving horizontally. Accept sin θ or cos θ with the 20.

cos θ = $\frac{1}{2}$

A1: Correct equation.

M1A1

θ = 60°

A1: Correct angle.

Accept $\frac{\pi}{3}$ or 1.05 (radians).

Allow 59.9 or better if they find W first

A1

3

(b) (W =)20 sin 60°

= 17.3 N

Or

(W =)$\sqrt{20^2 - 10^2}$ = 17.3 N

M1: Resolving vertically. Accept sin θ or cos θ with the 20, where θ is their answer to part (a) or 90 minus their answer to part (a).

A1: Correct weight CSO

or

M1: Correct use of Pythagoras eg $10^2 + W^2 = 20^2$

A1: Correct weight CSO

Accept $10\sqrt{3}$ *or AWRT 17.3*

M1A1F

2

(c) $m = \dfrac{20\sin 60°}{9.8} = 1.77$ kg

M1: Their answer to part (b) divided by 9.8.

A1F: Correct mass. Follow through their answer to part (b).
Accept 1.76 or 1.8.
Accept 2 sig figs in follow through.
Note: Using g = 9.81 gives the answer
1.77, also accept 1.76.

M1A1F

2

[7]

Q13.

$20g - T = 20a$

M1: Three term equation of motion for one particle.

M1

$T - 5g = 5a$

$15\,g = 25a$

M1: Three term equation of motion for the other particle.

A1: Both equations correct.

M1A1

$a = \dfrac{15g}{25} = 5.88$ ms^{-2}

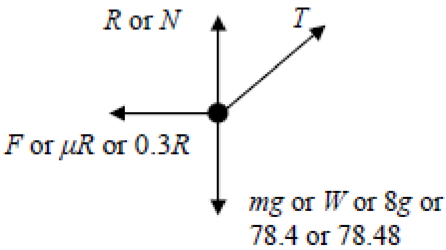
A1: Correct acceleration from correct working.

A1

[4]

Q14.

(a)



*B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.*

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 8 kg for the weight.

Note Accept μR or 0.3R for F.

B1

1

(b) R + 40 sin θ = 8 × 9.8

M1: Resolving vertically to obtain a three term equation, with R, T sin or cos(30° or 60°) and 8g oe.

A1: Correct equation

M1A1

R = 78.4 – 40sin θ

A1: Correct expression for R.

Accept (R=)8g – T sin30°

Note if using g = 9.81 accept

R = 78.48 – 0.5T or R = 78.5 – 0.5T

A1

3

(c) 8a = 40 cos θ – 0.3(78.4 – 40 sin θ)

M1: Use of the friction inequality with R from part (b)

A1: Correct equation.

M1A1

8a = – 23.52 + 40 cos θ + 12 sin θ

M1: Use of equation of motion with 40 sin θ or 40 cosθ and 8a and a friction term.

M1

p = – 2.94

q = 8

r = 1.5

A1: One correct value.

A1

A1: All three values correct.

A1

5

[9]

Q15.

(a) $20 \cos \theta = 10$

M1: Resolving horizontally. Accept sin θ or cos θ with the 20.

A1: Correct equation.

M1A1

$\cos \theta = \frac{1}{2}$

$\theta = 60°$

A1: Correct angle.

Accept $\frac{\pi}{3}$ *or 1.05 (radians).*

Allow 59.9 or better if they find W *first*

A1

3

(b) $(W =)\ 20 \sin 60°$

M1: Resolving vertically. Accept sin θ or cos θ with the 20, where θ is their answer to part (a) or 90 minus their answer to part (a).

M1

= 17.3 N

A1: Correct weight CSO

A1

Or

$(W =)\sqrt{20^2 - 10^2}$ = 17.3 N

M1: Correct use of Pythagoras eg $10^2 + W^2 = 20^2$

(M1)

A1: Correct weight CSO

Accept $10\sqrt{3}$ *or AWRT 17.3*

(A1)

2

(c) $m = \frac{20 \sin 60°}{9.8}$

M1: Their answer to part (b) divided by 9.8.

M1

= 1.77 kg

A1F: Correct mass. Follow through their answer to part (b).

Accept 1.76 or 1.8.

Accept 2 sig figs in follow through.

Note: Using g = 9.81 gives the answer 1.77, also accept 1.76.

A1F

2

[7]

Q16.

(a) $18g - T = 18a$

M1: Three term equation of motion for the 18 kg particle.

A1: Correct equation of motion for the 18 kg particle. (Accept $T - 18g = 18a$)

M1A1

$T = 12a$

B1: Equation of motion for the block that has signs consistent with the first equation.

B1

$18g - 12a = 18a$

$a = \frac{18g}{30} = 5.88\ \text{ms}^{-2}$

A1: Correct acceleration from correct work. Accept $\frac{3g}{5}$

Do not penalise consistent use of negative acceleration, provided final answer positive.

Special Case:

Whole String Method 18g = 30a and $a = \frac{18g}{30}$ = 5.88 OE M1A1

Note using g = 9.81 gives 5.89, also accept 5.88.

A1

4

(b) (i) $18g - T = 18 \times 3$

M1: Three term equation of motion for the 18 kg particle *with a = 3 seen.*

A1: Correct equation.

M1A1

$T = 18g - 18 \times 3 = 122(.4)\text{N}$

A1: Correct tension. Accept 122.4.

Note using g = 9.81 gives 123, also accept 122.

A1

3

(ii) $(R = 12 \times 9.8 = 117.6 \text{ N } =)118 \text{ N(to 3sf)}$

B1: Correct normal reaction. Accept 117 and 117.6. Final answer must be positive.

Do not accept 12g.

Note using g = 9.81 gives 118, also accept 117.

B1

1

(iii) $122.4 - F = 12 \times 3$

M1: Three term equation of motion for the block, containing their tension, F and 12 × 3.

A1F: Correct equation. Follow through T from part (b)(i).

M1A1F

$F = 86.4$

A1F: Candidate's T minus 36.

A1F

$86.4 = \mu \times 117.6$

dM1: Use of $F = \mu R$ with AWRT 117 or 118 for R and the candidate's value of F provided positive.

dM1

$\mu = \dfrac{86.4}{117.6} = 0.735$

A1: Correct μ. Accept anything between 0.728 and 0.739 inclusive. Allow 0.73 and 0.74.

A1

5

(c) No air resistance or no other forces

B1: One assumption from list

B1

Horizontal String

Block is a particle or **they** are particles

B1: For another assumption from list.

Do not penalise assumptions not in the list.

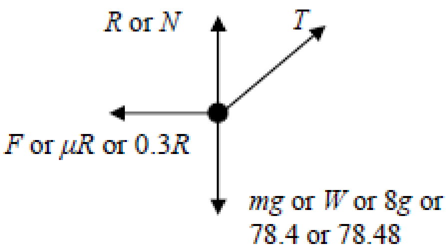
B1

2

[15]

Q17.

(a)



*B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.*

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 8 kg for the weight.

Note Accept μR or $0.3R$ for F.

B1

1

(b) $R + T \sin 30° = 8 \times 9.8$

M1: Resolving vertically to obtain a three term equation, with R, T sin or cos(30° or 60°) and 8 g oe.

A1: Correct equation

M1A1

$(R =)\ 78.4 - T \sin 30°$

$(R =)\ 78.4 - 0.5\ T$

A1: Correct expression for R.

Accept $(R =)\ 8g - T \sin 30°$

Note if using $g = 9.81$ accept $R = 78.48 - 0.5T$ or $R = 78.5 - 0.5T$

A1

3

(c) $T\cos 30° - F = 8 \times 0.05$

M1: Horizontal equation of motion with F, T sin or cos (30° or 60°) and 8×0.05 oe.

A1: Correct equation.

M1A1

$F = 0.3(78.4 - \sin 30°)$

M1: Using $F = 0.3R$ with their R from part (b), provided it includes a term in T.

A1: Correct expression for friction.

M1A1

$T\cos 30° - 0.3(78.4 - T\sin 30°) = 0.4$

$T = \dfrac{23.52 + 0.4}{\cos 30° + 0.3\sin 30°} = 23.5\text{N}$

dM1: Solving for T. Must see (cos 30° ± 0.3sin 30°) or similar in the denominator. (Dependent on both previous M marks.)

A1: Correct T. Accept 23.6 or AWRT 23.5

dM1A1 6

Or

$T\cos 30° - F = 8 \times 0.05$

M1: Horizontal equation of motion with F, T sin or cos(30° or 60°) and 8 × 0.05 oe.

A1: Correct equation.

(M1A1)

$T\cos 30° - 0.3R = 8 \times 0.05$

M1: Using $F = 0.3R$

A1: Two correct equations involving only T and R.

(M1A1)

$R + T\sin 30° = 8 \times 9.8$

solving simultaneously gives

$T = 23.5$

dM1: Solving for T.

A1: Correct T. Accept 23.6 or AWRT 23.5

Note using g = 9.81 gives 23.6, also accept 23.5.

(dM1A1)

[10]

Q18.

(a) Using $F = ma$:

$m\frac{\mathrm{d}v}{\mathrm{d}t} = 49 - 9.8v$ or $5g$ 9.8 □v

Need to see $m\frac{\mathrm{d}v}{\mathrm{d}t}$ *or* $5\frac{\mathrm{d}v}{\mathrm{d}t}$ *or* $a = \frac{49 - 9.81}{5}$

M1

$\therefore \frac{\mathrm{d}v}{\mathrm{d}t} = -1.96(v - 5)$

Must see m *terms (not* a *=…)*

A1

2

(b) $\int \frac{dv}{v-5} = -1.96\int \mathrm{d}t$

And one side integrated

M1

$\ln(v - 5) = -1.96t + c$

Need $+ c$*, A1 each side*

A1A1

When $t = 0$, $v = 7 \Rightarrow c = \ln 2$

OE

A1

$\ln\frac{v-5}{2} = -1.96t$

$\frac{v-5}{2} = e^{-1.96t}$

$v = 5 + 2e^{-1.96t}$

CAO

A1

5

[7]

Q19.

(a) $T = 2 \;\; 9.8 = 19.6$ N

M1: Equating tension and weight.

A1: Correct tension CAO

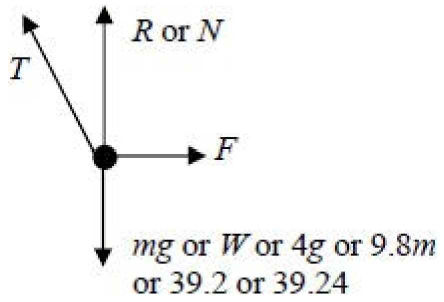
Accept $2g$

Accept 19.62 from $g = 9.81$

M1A1

2

(b)



B1: R, F (not μR) and mg correct

B1

B1: T correct, must be in roughly correct direction.

If more than four forces shown, do not award more than one mark.

Note all forces must be shown as arrows and have labels.

Note some candidates may draw the force diagram in the section with the question.

Components can be ignored if shown in a different notation eg dashed arrows.

B1

2

(c) $T \cos 30° + R = 4 \;\; 9.8$

M1: Resolving vertically to form a three term equation. (May be implied)

M1

(R =) 39.2 – 19.6cos30°
= 39.2 – 16.9741...

A1 Correct expression for R or equation for R.
Must see 19.6cos30 or equivalent (eg $2g$sin60)

A1

= 22.2259....
= 22.2 N (to 3sf) AG

A1: Correct force. Must see intermediate working, for example third or fourth line of working in solution opposite.

Example: 19.6sin 30° – R = 4 9.8 scores M1A0A0.

Use of g = 9.81 still gives 22.2 N as the final answer.

A1

3

(d) $T\cos 60° = F$

M1: Resolving horizontally

M1

$F = 19.6\cos 60° = 9.8$

A1: Correct expression for friction

A1

$19.6\cos 60° \le \mu(39.2 - 19.6\cos 30°)$

dM1: Use of $F = \mu R$ or $F \le \mu R$ (do not allow $F \ge \mu R$)

dM1

$$\mu \ge \frac{19.6\cos 60°}{39.2 - 19.6\cos 30°}$$

$\mu \ge 0.441$

A1: Final answer of μ = 0.441 or $\mu \ge 0.441$ from correct working

Use of g = 9.81 still gives 0.441 as the final answer.

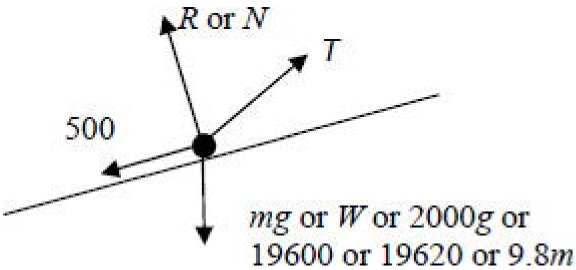
A1

4

[11]

Q20.

(a)



B1: R, 500 and mg correct

B1

B1: Tension in roughly correct direction.

If more than four forces shown, do not award more than one mark.

Note all forces must be shown as arrows and have labels.

Note some candidates may draw the force diagram in the section with the question.

Components can be ignored if shown in a different notation eg dashed arrows.

B1

2

(b) 2000 0.6 = T cos12° – 500 – 2000 9.8sin5°

M1: Resolving parallel to the slope to obtain a four term equation of motion. The weight and tension terms must be resolved.

A1: Correct terms.

A1: Correct signs.

M1A1A1

$T = \dfrac{1200 + 500 + 19600 sin 5°}{cos 12°}$

$\left(= \dfrac{3408.25}{cos 12°}\right)$

(= 3484.4)

dM1: Solving for T.

dM1

= 3480 (to 3sf) AG

A1: Correct tension. AWRT 3480.

Allow AWRT 3490 from use of g = 9.81.

A1

5

[7]

Q21.

(a) (i) $0.6^2 = 0^2 + 2a \times 0.9$

M1: Correct use of constant acceleration equation with u = 0 to find a.

A1: Correct equation.

M1A1

$a = \frac{0.6^2}{1.8} = 0.2 \text{ ms}^{-2}$ **AG**

A1: Correct a but some intermediate working must be seen.

A1

Note that $0^2 = 0.6^2 + 2a \times 0.9$

Scores M0A0A0

Verification methods require a conclusion for full marks to be awarded.

Condone seeing just the second line of working.

3

(ii) $0.9 = \frac{1}{2}(0 + 0.6)t$

M1: Correct use of constant acceleration equation
with $u = 0$ (and $a = 0.2$ if needed) to find t.

M1

$t = \frac{0.9}{0.3} = 3$ seconds

A1: Correct time.

A1

Note: Do not penalise $0.9 = \frac{1}{2}(0.6 + 0)t$ in the first method.

OR

$0.6 = 0 + 0.2t$

(M1)

$t = \frac{0.6}{0.2} = 3$ seconds

(A1)

Note: $0 = 0.6 + 0.2t$ scores M0A0 in the second method.

OR

$0.9 = \frac{1}{2} 0.2t^2$

(M1)

$t = 3$ seconds

(A1)

2

(b) $T - 800 \times 9.8 = 800 \times 0.2$

M1: Three term equation of motion. Must have these three terms but can have incorrect signs. Must use $a = 0.2$

A1: Correct equation with correct signs.

(Allow $800g$)

M1A1

$T = 7840 + 160 = 8000$ N

A1: Correct tension.

A1

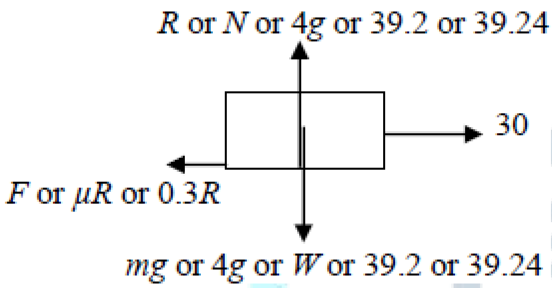
Accept 8008 or 8010 from use of $g = 9.81$.

3

[8]

Q22.

(a)



B1: Diagram with four forces showing arrow heads and labelled.

Ignore negative signs in labels.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 4 kg for the weight.

Note Accept μR for F.

B1

1

(b) ($R = 4 \times 9.8 =$) 39.2 N

B1: Correct normal reaction. Accept $4g$

B1

1

(c) ($F =$)$0.3 \times 39.2 = 11.76 = 11.8$ N (to 3sf)

M1: Use of ($F =$)μR

M1

A1: Correct friction.

Accept $1.2g$ or 11.7 or 11.76 N.

Do not condone further work after the value for friction has been obtained.

A1

2

(d) $4a = 30 - 11.76$

M1: Three term equation of motion.
A1F: Correct equation.

M1A1F

$$a = \frac{30 - 11.76}{4} = 4.56 \text{ ms}^{-2}$$

A1F: Correct acceleration.
FT candidates F from part (c).
Accept 4.55 from 11.8.

A1F

3

[7]

Q23.

(a) $5g - T = 5a$

M1: Three term equation of motion with 5g or 49, 5a (not 5ga) and T.
A1: Correct equation.

M1A1

$T - 3g = 3a$

M1: Three term equation of motion with 3g or 29.4, 3a (not 3ga) and T.
A1: Correct equation.

M1A1

$2g = 8a$

$$a\left(= \frac{2g}{8}\right) = 2.45 \text{ ms}^{-2}$$

AG

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they then give their final answer as 2.45. If the final answer is –2.45 don't award the final A1 mark.

Special Case:

Whole String Method $2g = 8a$ and $a = \frac{2g}{8} = 2.45$ *OE M1A1A1.*

5

(b) $T = 3 \times 9.8 + 3 \times 2.45$

M1: Substitution of a = 2.45 into a three term equation of motion to find the tension. Contains T, mg and ma where m = 3 or 5

M1

= 36.75

= 36.8 N (to 3 sf)

A1: Correct tension.
Accept 36.75 or 36.7

A1

2

(c) Light and Inextensible

B1: Light
B1: Inextensible (Allow inelastic or not stretchy)
Ignore irrelevant non-contradictory assumptions.

B1B1

2

(d) (i) $0.196 = \frac{1}{2} \times 2.45 \times t^2$

M1: Use of constant acceleration equation with s = 0.196, u = 0 and a = 2.45 to find t.
A1: Correct equation.

M1 A1

$$t = \sqrt{\frac{2 \times 0.196}{2.45}} = 0.4 \text{ seconds}$$

A1: Correct t

A1

3

(ii) $v^2 = 0^2 + 2 \times 2.45 \times 0.196$

$v = 0.98$

M1: Use of constant acceleration equation with s = 0.196, a = 2.45, u = 0 and candidate's time (as needed) to find v.
A1: Correct v.

M1A1

OR

$v = 0 + 2.45 \times 0.4 = 0.98\ \text{ms}^{-1}$

(M1A1)

OR

$$0.96 = \frac{1}{2}(0 + v) \times 0.4$$

(M1)

$v = 0.98\ \text{ms}^{-1}$

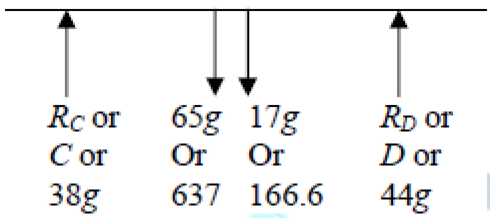
(A1)

2

[14]

Q24.

(a)



B1: Two weights correct and in correct relative positions.

B1: Two upward reaction forces, labelled differently.

Note all forces must be shown as arrows and have labels.

Condone use of g = 9.81 for calculating weights.

B1
B1

2

(b) Taking moments about C

$3 \times 17g + 2.6 \times 65g = 44g \times d$

B1: Seeing 2.6.

M1: Three term moment equation including $17g$, $65g$ and $44g$ or 17, 65 and 44, with different distances for the $17g$ and $65g$.

A1: Correct equation.

B1
M1
A1

$44d = 220$

$d = 5$

Distance is $5 - 4.6 = 0.4$ m

4

A1: Correct final answer.

A1

Alternative

$R_C = 38g$

(B1)

Taking moments about D

Could take moments about any other point

$38g(4.6 + x) = 65g(2 + x)+17g(1.6 + x)$

(M1)
(A1)

$174.8 – 130 – 27.2 = 44x$

$x = 0.4$

(A1)

(4)

(c) Gravitational force (centre of mass or weight) at mid-point (or centre) of the plank

E1

E1: Correct explanation.

1

[7]

Q25.

(a) $0 = 4.9 – 9.8t$

M1: Use of constant acceleration equation to find t with v = 0.
A1: Correct equation

M1A1

$t = \frac{4.9}{9.8} = 0.5\text{ s}$

A1: Correct time

A1

3

(b) $0^2 = 4.9^2 – 2 \times 9.8s$

M1: Use of constant acceleration equation to find s with v = 0.

M1

$s = \frac{4.9^2}{19.6} = 1.225\text{ m}$

A1: Correct height.

A1

H_{max} = 1.225 + 1 = 2.23 m

A1: Correct total height.

A1

3

(c) 1 second

B1: Correct time.

B1

1

(d) 4.9 m s^{-1}

B1: Correct speed.

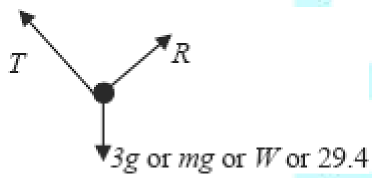
B1

1

[8]

Q26.

(a)



B1: Correct force diagram with arrows and sensible labels.

If R is shown as vertical award B0.

If F is included, award B0

Accept a reflection of the diagram in a vertical line.

Ignore components if shown with a different notation, eg dotted lines.

B1

1

(b) (R =) 3g cos 60

M1: Resolving perpendicular to the slope. Must see cos 60° or sin 30° or cos 30° or sin 60° and 3g or 29.4.

NOTE: $\frac{3g}{2} = 14.7$ *or equivalent without the use of a trig term scores M0.*

M1

(R =) 14.7 **AG**

A1: Correct value from correct working.
NOTE: If candidates use g = 9.81, deduct one mark here. If candidates obtain 14.7 from 14.715 they will have used g = 9.81.
Note: "R =" does not need to be seen.

A1

2

(c) (T =) 3g sin 60°

M1: Resolving parallel to the slope. Must see cos 60° or sin 30° or cos 30° or sin 60° and 3g or 29.4.

M1

(T =) 25.5

A1: Correct value. AWRT 25.5 or truncation to 25.4.
NOTE: If candidates use g = 9.81 again, do not penalise. Use of g = 9.81 gives 25.5 for the tension.
Note: "T =" does not need to be seen.

A1

2

[5]

Q27.

(a) $v^2 = 0^2 + 2 \times 9.8 \times 15$

M1: Use of constant acceleration equation to find v with u = 0 and a = ± 9.8.

M1

$v^2 = 294$

A1: Correct equation

A1

v = 17.1 m s^{-1}

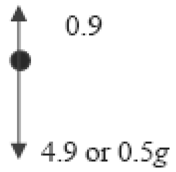
A1: Correct speed from correct working. Accept AWRT 17.1. Accept 17.15.
Accept $7\sqrt{6}$
Note: If g = 9.81 is used for the first time deduct one mark. Should get 17.2 m s^{-1}

from g = 9.81.

A1

3

(b) (i)



B1: Correct diagram, with arrows and labels. Must see 0.9 and 4.9 or 0.5g (or 4.905 if working with g = 9.81).

B1

1

(ii) $4.9 - 0.9 = 0.5a$

M1: Uses 0.5a.
B1: Explicit statement of "4.9 – 0.9" or "mg – 0.9" or "0.5g – 0.9".

M1B1

$(a =)\dfrac{4}{0.5} = 8$ ms^{-2} **AG**

A1: Correct acceleration from correct working. Can be awarded without the B1 mark.

Must see $\dfrac{4.9(\text{or } 0.5g) - 0.9}{0.5}$ or $\dfrac{4}{0.5}$ *or 4 = 0.5a*

A1

Note: If g = 9.81 is used candidates will get 8.01 m s^{-2}. Deduct 1 mark if 8.01 is seen.
Examples:
4.9 = 0.5a + 0.9 M1 B0 A0
a = 8
4 = 0.5a M1 B0 A1
a = 8
If candidates only write

$a = \dfrac{4}{0.5} = 8$ *award M0 B0 A0.*

3

(iii) $v^2 = 0^2 + 2 \times 8 \times 15$

M1: Use of constant acceleration equation to find v with u = 0 and a = ± 8.

M1

v = 15.5 m s^{-1}

A1: Correct speed from correct working.
Accept AWRT 15.5 or truncated to 15.4.

Accept $4\sqrt{15}$.

A1

2

(iv) The air resistance force will not be constant, but changes **as the speed of the ball changes** (or **changes as the ball accelerates**).

B1: Correct explanation, key words in bold.
Do not award mark for statements that imply that the acceleration causes the air resistance to change.

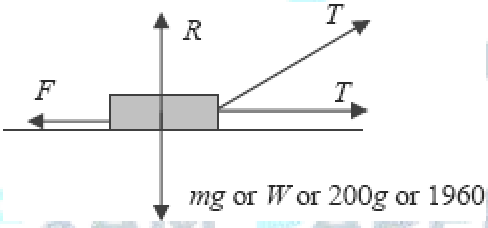
B1

1

[10]

Q28.

(a)



B1: F, R and mg (or equivalent) with arrows and labels.

B1

B1: Two ***equal*** *tension forces with arrows and labels.*

B1

Ignore components if shown with a different notation, eg dotted lines.

2

(b) $R + T \sin 20° = 1960$

OR

$R + T \sin 20° = 200g$

M1: Resolving vertically with three terms.
Must include sin 20° or cos 20° or sin 70° or cos 70° with T and 200g or 1960.

A1: Correct equation.

M1A1

(R =) 1960 – T sin 20°
OR
(R =) 200g – T sin 20°

A1: Correct expression for the normal reaction.
Note: If g = 9.81 is used for the first time deduct one mark. Should get 1962 instead of 1960

A1

3

(c) $T \cos 20° + T - F = 200 \times 0.3$

M1: Four term equation of motion. Must include sin 20° or cos 20° or sin 70° or cos 70° with T and a second T term with no trig.
A1: Correct equation

M1A1

$T \cos 20° + T - 0.4\,(1960 - T \sin 20°) = 200 \times 0.3$

M1: Use of friction law with their expression for R, provided that R has two terms. Note that this mark does not depend on any previous marks.
Example:
If Candidate gives 1960 as answer to part (b), then: F = 0.4 × 1960 = 784 scores M0 here

M1

$$T = \frac{60 + 784}{\cos 20° + 1 + 0.4 \sin 20°} = 406$$

dM1: Solving for T.

dM1

Note: This mark requires both of the previous M marks.
A1: Correct tension.
Accept AWFW 406 to 407.
Note: If g = 9.81 is used should get 407 instead of 406.

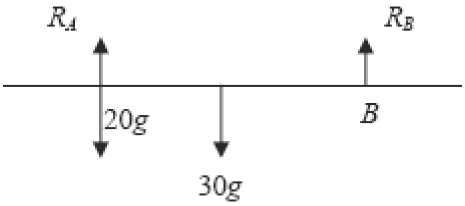
A1

5

[10]

Q29.

(a)



B1 for four forces
B2 for two different reactions and 30g and 20g marked

B2 **2**

(b) Taking moments about A:
$3.2 \times 30g = R_B \times 5$
B1 for 3.2

M1B1

$R_B = 19.2g$
AG

A1 **3**

(c) Resolve vertically: $R_A + R_B = 50g$
Can be awarded in (b)

M1

$R_A = 30.8g$ or 302 N

A1 **2**

(d) Gravitational force acts through mid-point of the rod

E1 **1**

[8]

Q30.

(a) $12g - T = 12a$
M1: Three term equation of motion, with 12g (or 117.6), 12a (not 12ga) and T.
A1: Correct equation

M1A1

$T - 8g = 8a$

M1: Three term equation of motion, with $8g$ (or 78.4), $8a$ (not $8ga$) and T.
A1: Correct equation

M1A1

$4g = 20a$

$a\left(= \frac{4g}{20}\right) = 1.96 \text{ m s}^{-2}$ AG

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 1.96. If final answer is – 1.96 don't award final A1 mark.

Special Case:

Whole String Method $4g$ = $20a$ = and

$a = \frac{4g}{20} = 1.96$ *OE M1A1A1*

5

(b) $T = 8g + 8 \times 1.96 = 94.1$ N

M1: Use of three term equation of motion to find T, with a = 1.96.
A1: Correct tension.
Accept 94.08.

M1A1

2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v, with a = 1.96 and u = 0.

A1: Correct v.

Using s = 4 scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation to find v, with a = ±9.8 and u ≠ 0.

M1

A1F: Correct equation. FT initial velocity from (c)(i).

A1F

v = 9.68 m s^{-1}

A1F: Correct v. FT initial velocity from (c)(i).
For example 11.8 from 7.84.

A1F

3

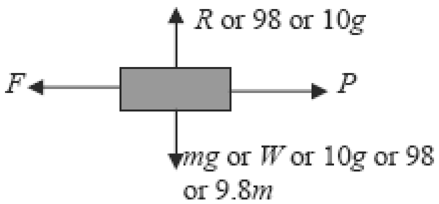
Use of g = 9.81

(a) 1.962	*M1A1M1A1A0*
(b) 94.2	*M1A1*
(c) (ii) 9.69	*M1A1A1*

[12]

Q31.

(a)



B1: Correct force diagram with arrows and labels.

Note: Award mark if forces drawn on the diagram in the question.
Note: Do not accept 10 kg for the weight.
Note: Do not accept μR or 0.5R for F.

B1

1

(b) (i) (R = 10 × 9.8 =) 98 N

B1: Correct normal reaction. Accept 10g.
No need to see the letter R or working.

B1

1

(ii) (F ≤) 0.5 × 98

(F ≤) 49

B1: Correct maximum value for friction.
Accept 5g.
No need to see the letter F or any working. Ignore any inequalities.
For FT, must be 0.5 of candidate's answer to (b)(i).

B1F

1

(iii) (F =) 30 N

B1: Correct friction. Allow – 30.

B1

1

(c) $80 - 49 = 10a$

M1: Three term equation motion, containing 80, candidate's 49 and 10a (not 10ga) in any combination.
A1F: Correct equation including signs.

M1A1F

$a = 3.1 \text{ m s}^{-2}$

A1F: Correct acceleration.
FT candidate's answer to (b)(ii).

A1F

3

Allow use of g = 9.81

(b) (i) 98.1	*B1*
(b) (ii) 49.05 or 49.1 or 49	*B1*
(c) 3.095 or 3.09 or 3.1	*M1A1A1*

[7]

Q32.

(a) $12g - T = 12a$

M1: Three term equation of motion, with 12g (or 117.6), 12a (not 12ga) and T.
A1: Correct equation

M1A1

$T - 8g = 8a$

M1: Three term equation of motion, with 8g (or 78.4), 8a (not 8ga) and T.
A1: Correct equation

M1A1

$4g = 20a$

$a\left(=\frac{4g}{20}\right) = 1.96 \text{ m s}^{-2}$ AG

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they *give their final answer as 1.96. If final answer is – 1.96 don't award final A1 mark.*

Special Case:

Whole String Method $4g = 20a$ and

$a = \dfrac{4g}{20} = 1.96$ *OE M1A1A1*

5

(b) $T = 8g + 8 \times 1.96 = 94.1$ N

M1: Use of three term equation of motion to find T, with $a = 1.96$.

A1: Correct tension.

Accept 94.08.

M1A1

2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92$ m s^{-1}

M1: Use of constant acceleration equation to find v, with $a = 1.96$ and $u = 0$.

A1: Correct v.

Using $s = 4$ scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation to find v, with $a = \pm 9.8$ and $u \neq 0$.

M1

A1F: Correct equation. FT initial velocity from (c)(i).

A1F

$v = 9.68$ m s^{-1}

A1F: Correct v. FT initial velocity from (c)(i). For example 11.8 from 7.84.

A1F

3

(iii) $4 = \dfrac{1}{2}(-3.92 + 9.68)t$

M1: Use of $s = \dfrac{1}{2}(u + v)t$

M1

A1: Correct values.

A1

A1: Correct signs.

A1

t = 1.39

dM1: Solving for t.

dM1

A1: Correct t.

A1

OR

$$-4 = 3.92t - 4.9t^2$$

$$4.9t^2 - 3.92t - 4 = 0$$

$$t = \frac{3.92 \pm \sqrt{3.92^2 - 4 \times 4.9 \times (-4)}}{2 \times 4.9}$$

M1: Forming a quadratic with candidates u from (c)(i) or v from (c)(ii) with 4.9 or 9.8.

(M1)

A1: Correct terms in quadratic.

(A1)

A1: Correct signs in quadratic.

(A1)

t = 1.39 or t = –0.588

dM1: Solving quadratic (do not penalise for negative discriminant).

(dM1)

t = 1.39

A1: Correct root seen (other root does not need to be seen).

(A1)

OR

$t_{up} + t_{down}$ = 0.4 + 0.4 + 0.588

M1: Finding total time from two or three times.

(M1)

A1: 0.4 or 0.8 seen.

(A1)

dM1: Finding second or third time for

downward motion.

(dM1)

A1: Obtaining 0.588 or 0.988.

(A1)

= 1.39 (to 3SF)

A1: 1.39. Accept 1.38.

(A1)

OR

9.68 = –3.92 + 9.8t

M1: Use of $v = u + at$

(M1)

A1: Correct values.

(A1)

A1: Correct signs.

(A1)

$t = \frac{13.6}{9.8} = 1.39$

dM1: Solving for t

(dM1)

A1: Correct t

(A1)

5

Use of g *= 9.81*

(a) 1.962	*M1A1M1A1A0*
(b) 94.2	*M1A1*
(c) (ii) 9.69	*M1A1A1*
(c) (iii) 1.39	*M1A1A1dM1A1*

[17]

Q33.

(a) $11g - T = 11a$

M1: Equation of motion for A*, containing* T*, 11*g *or 107.8 and 11*a*.*

M1

A1: Correct equation

A1

$T - 9g = 9a$

M1: Equation of motion for B containing T, $9g$ or 88.2 and $9a$.

M1

A1: Correct equation

A1

$2g = 20a$

AG

$a = 0.98 \text{ m s}^{-2}$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 0.98. If final answer is – 0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and

$a = 2g / 20 = 0.98$ OE M1A1A1

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

(b) (i) $v = 0 + 0.98 \times 0.5 = 0.49 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225$ m

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$0.49^2 = 0^2 + 2 \times 0.98s$

M1: Finding distance travelled by each particle with u = 0, a = 0.98 and their v.

(M1)

$$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$$d = 2 \times 0.1225$$

M1: Doubling distance or use of d/2 in their original equation.

M1

= 0.245 m

A1: Correct final distance. Allow 0.244 or 0.246. (Use of 0.5 × 0.49 = 0.245 scores zero unless justified)

A1

4

[11]

Q34.

(a) $4a = 4g \sin 40°$

M1: Resolving and application of Newton's second law. Allow cos 40°

M1

A1: Correct expression.

A1

$a = g \sin 40° = 6.30$ m s^{-2} **AG**

A1: Correct result from correct working. Must see 6.30 not 6.3.

Just seeing g sin 40° = 6.30 m s^{-2} scores full marks.

Use of g = 9.81 gives 6.31, M1A1A0, but don't penalise again on the same script.

A1

3

(b) $$0.9 = 0 + \frac{1}{2} \times a \times 0.6^2$$

M1: Use of a constant acceleration equation to find a, with s = 0.9, u = 0 and = 0.6.

M1

A1: Correct equation

A1

$$a = \frac{0.9 \times 2}{0.6^2} = 5 \text{ m s}^{-2}$$

A1: Correct acceleration

A1

ALT Method

$$09.9 = \frac{1}{2}(0 + v) \times 0.6$$

v =3

No marks at this stage.

3 = 0 + 0.6a

M1: Constant acceleration equation with u = 0 and t = 0.6.

(M1A1)

A1: Correct equation

a = 5 m s^{-2}

A1: Correct acceleration.

(A1)

3

(c) The acceleration is reduced because of air resistance or the fact that there is friction.

B1: Must mention air resistance/resistive forces or friction. Do not allow air friction.

B1

1

[7]

Q35.

(a) Peg is smooth

B1: Correct assumption

B1

1

(b) String is light

B1: First correct assumption

B1

String is inextensible or inelastic

B1: Second correct assumption

B1

Tension is the same throughout the string

Note: Ignore any additional assumptions.

2

(c) $11g - T = 11a$

M1: Equation of motion for A, containing, $11g$ or 107.8 and $11a$.

M1

A1: Correct equation

A1

$T - 9g = 9a$

M1: Equation of motion for B containing, $9g$ or 88.2 and $9a$.

M1

A1: Correct equation

A1

$2g = 20a$

AG

$a = 0.98 \text{ m s}^{-2}$

A1: Correct acceleration from correct orking.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 0.98. If final answer is – 0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and

$a = 2g/20 = 0.98$ OE M1A1A1

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

(d) (i) $v = 0 + 0.98 \times 0.5 = 0.49$ m s^{-1}

M1: Use of constant acceleration equation to find v with u = 0, a = 0.98 and t = 0.5.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225$ m

M1: Finding distance travelled by each particle with u = 0, a = 0.98 and t = 0.5.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$0.49^2 = 0^2 + 2 \times 0.98s$

M1: Finding distance travelled by each particle with u = 0, a = 0.98 and their v

(M1)

$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$d = 2 \times 0.1225$

M1: Doubling distance or use of d/2 in their original equation.

M1

= 0.245 m

A1: Correct final distance. Allow 0.244 or 0.246. (Use of 0.5 × 0.49 = 0.245 scores zero unless justified)

A1

4

If candidates calculate the distance first award marks as above (see (d) (i)) or:

M1: Use of constant acceleration equation

to find v with u = 0, a = 0.98 and s = 0.1225.
A1: Correct v
Note: If parts (i) and (ii) are not separated or clearly labelled still award marks for both parts if justified.

[14]